



NATIONAL SENIOR CERTIFICATE EXAMINATION
NOVEMBER 2010

INFORMATION TECHNOLOGY: PAPER I

MARKING GUIDELINES

Time: 3 hours

180 marks

These marking guidelines were used as the basis for the official IEB marking session. They were prepared for use by examiners and sub-examiners, all of whom were required to attend a rigorous standardisation meeting to ensure that the guidelines were consistently and fairly interpreted and applied in the marking of candidates' scripts.

At standardisation meetings, decisions are taken regarding the allocation of marks in the interests of fairness to all candidates in the context of an entirely summative assessment.

The IEB will not enter into any discussions or correspondence about any marking guidelines. It is acknowledged that there may be different views about some matters of emphasis or detail in the guidelines, and different interpretations of the application thereof. Hence, the specific mark allocations have been omitted.

QUESTION 1

- 1.1 A signal on one circuit/channel interferes with a signal on another circuit/ channel electromagnetic interference (between two wires/circuits) that disrupts communication.
- 1.2 A disk file system
- 1.3 A malicious program that disguises itself as another useful and innocent program.
- 1.4 An operating system used in mobile devices. Accept "Android is an OS developed and implemented by Google."
- 1.5 A keylogger is a program or a hardware device which detects and records a user's keystrokes/mouse actions.
- 1.6 A computer on a network with reduced cpu, memory, storage (anything that indicates less power/cheaper).
- 1.7 Software where everyone has access to the source code.
- 1.8 Radio frequency identification system used for embedding an ID chip in stock, pets, etc.
- 1.9 2 physical CPUs are included on a single chip.
- 1.10 Dynamically generated Web content which facilitates interactive information sharing.

[20]

Question 2

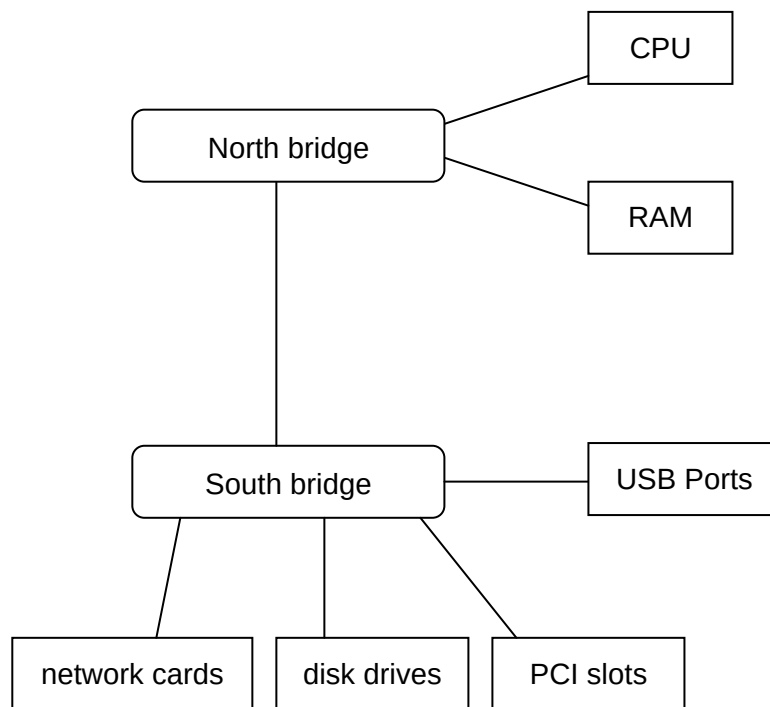
2.1

NB: CPU & USB switch.

- Correct diagram with each switch correct labels
- Errors – 1 each

Correct diagram:

(4)



2.2

2.2.1 A set of instructions which a CPU can execute. (2)

2.2.2 Breaks instructions into stages like an assembly line. Works on multiple stages at once. (2)

2.2.3 RISC uses instructions of the same length, pipelining prefers instruction of the equal length thus it is optimal for all instruction lengths to be the same. Thus RISC is suitable for pipelining. (2)

2.3 fetch, decode, execute and data transfer/write back (4)

2.4

2.4.1 one of:

data to be processed by CPU
 instruction to be read
 result of calculation/comparison
 instruction counter
 memory address

(1)

2.4.2 cache (1)

2.5

- 2.5.1 Any THREE:
Smaller transistors = more in same space = more processing power.
Smaller also = less heat and greater efficiency. (3)
- 2.5.2 In a single program more than one thread ('task') is carried out at the same time. (2)
- 2.5.3 Two or more processors function together within one computer system. (2)
- [23]**

QUESTION 3

3.1 Any TWO of:

- Perform the POST/test the hardware
 - Load the OS
 - Power management
 - Video initialisation
 - I/O calls
- (2)

3.2

3.2.1 the client computer gets its data from the proxy instead of the real server. (2)

3.2.2 Any TWO of:

- file server
 - print server
 - mail server
 - application server
 - FTP server
 - chat server
 - list server
- (2)

3.3 Any THREE of: at TWO marks each ($3 \times 2 = 6$)

- File compression program compresses a file to a smaller size
 - Backup utility – enables one to make backups
 - Disk cleanup frees up disk space by removing temporary and unused files
 - Checkdisk or scandisk searches a disk for physical errors or bad sectors
 - Disk defragmenter physically reorganises the location of file sections so that there is less fragmentation on the disk
- ... add other valid answers (6)

3.4 Printing is standardised by the operating system which uses drivers to communicate with the printers. OR

Any answer which indicates that the application prints to a driver and the OS rather than to a specific physical printer. (2)

3.5

3.5.1 Cloud computing refers to the use of applications and data which are hosted on servers on the Internet. (2)

3.5.2 Three points either in support **OR** opposition. (Both not acceptable)

Support:

- do not have to install/maintain software
- easy to share/collaborate online
- lower cost/free
- not dependent on a machine – can have a computer with low specifications
- not dependant on an OS/hardware architecture
- as long as you have access to the Internet somewhere, you have access to all your documents
- software automatically updated

Oppose:

- If no Internet access, then you have no access to your own documents
- Could lose all documents in case of theft/damage of Internet company's hardware or software
- Risk of hackers
- At risk of monthly service fees (software as a service)

(3)
[19]

QUESTION 4

4.1

Connection	Connection Technology	Physical Medium
A	UTP/STP	Copper wire
B	Fiber Optic	Light
C	Bluetooth	Radio waves
D	WiFi/Wireless network	Radio waves

(1 mark per cell)

(8)

4.2

4.2.1 A router links networks which use different protocols. (2)

4.2.2 A switch connects computers within one network; a router links different networks.
A switch cannot communicate with different protocols, whereas a router can. (2)

4.3

4.3.1 TCP/IP (1)

4.3.2 Any TWO of:
HTTP/FTP/POP3/SMTP/P2P/IMAP/HTTPS (others to be judged on merit) (2)

4.4 The wrong protocol is being used and unless they use TCP/IP they cannot connect. (2)

4.5

4.5.1 CSMA/CD (1)

4.5.2 Token ring/Token Passing (1)

4.6 Messages are broken up into variable length packets; these packets are sent from sender to receiver via several routers; packets can follow different routes to get to destination; the message is reassembled at the destination. (4)

[23]

QUESTION 5**5.1**

- 5.1.1 Voice over Internet Protocol/Voice over IP (1)
- 5.1.2 any THREE of:
- free
 - usually good sound quality
 - encrypted
 - using webcam callers can see each other (3)
- 5.2.1 **virus hoax** – Any TWO wastes time, clogs email, makes people panic and do stupid things, wastes bandwidth. (2)
- 5.2.2 **spam** – Any TWO wastes time, wastes bandwidth, embarrassing, invades privacy, clogs email. (2)
- 5.2.3 **trojan horse** – Any TWO steals data, destroys data, allows unauthorised access to computer, creates a 'backdoor', acts as spyware. (2)
- 5.2.4 **phishing** – Any THREE steals data, steals personal details, can cause loss of money, can lead to fraud, can lead to identity theft. (3)

5.3

- 5.3.1 Stealing someone's personal identifying information in order to impersonate that person. (2)
- 5.3.2 Any THREE keystrokes, web sites accessed, mouse movements, mouse clicks, software installed, machine specifications (3)
- 5.3.3 Through spyware a criminal can acquire one's username and password, and go online pretending to be the owner of this username and password. (2)
- 5.3.4 Any TWO of:
- She should change her password often
 - She should make use of a strong password
 - She should not use the same password on more than one site
 - She should never make her password known to anyone else
 - When browsing other sites on the Internet, she should not give out any personal information unless it is a trusted site (2)

- 5.4 Any THREE of:
She should check to see whether the site:
- contains links to other sources which appear to be reliable
 - has been updated recently
 - lists the authors
 - is owned by a government or educational agency
 - is clearly aimed at selling (then perhaps not trustworthy)
 - is free of grammar and spelling errors
 - has content which can be verified in the form of similar methods/advice on other sites
- (3)
- 5.5
- 5.5.1 encryption (1)
- 5.5.2 digital certificate (1)
- 5.6 Any TWO:
- She acknowledged the author of this text by supplying a reference.
 - She quoted the text in a context of criticism and/or education of others.
 - Does not pretend it is her own work.
- (2)
- [29]**

QUESTION 6

- 6.1 Any THREE that emphasise why business/schools dislike social networking (not an individual).
- Distracts from work
 - Lowers productivity
 - Increases bandwidth usage
 - People can put data out that compromises the school/business
 - Costs money
- (3)
- 6.2
- 6.2.1 Anything along the lines of:
- KidBook receives information from users. They must let the users know what they will do with this information/who has access to the information.
Candidates more likely to only answer:
 - They must let the users know what they will do with this information/who has access to the information.
- (2)
- 6.2.2 Any THREE. It stores data that allows a web site to recognise you, log you in, personalise itself, remember your settings and remember your last actions.
- (3)
- 6.2.3 Accept answers which have valid substantiations – for example:
- No, as a cookie is just a text file, it cannot contain a virus.
 - No, cookies only store information which is used by the browser.
 - Yes, cookies store personal information which can be passed on to third parties on the Internet.
- (2)

6.3

6.3.1 Allow any TWO of the following:

- It is written by anyone, not controlled by a publisher, is available on the Internet, is likely to be up to date, may have inaccurate information.

(2)

6.3.2 Any TWO of:

- Wiki content might not be reliable/not have facts checked
 - Anyone can edit the content
 - Content can be made up/opinion
 - Wiki pages are open to vandalism
 - Information on the Wiki could become disorganised/unstructured
- ... add other valid answers

(2)

6.4 Any TWO of:

- Jordan must ensure that Kidbook has electronic copies of all work that Christine did for the company.
- He must delete Christine's user account on the server.
- Delete her email account.
- He must have a backup of all email correspondence that Christine did on behalf of the company.

(2)

6.5 Marks are allocated on the Answer Sheet – a total of 6 marks for TWO careers for Sibongile; a total of 3 marks for ONE career for Adam.

Mark allocation:

3 marks per career, allocated as follows:

- one mark for chosen career if the reasons given are valid
- two marks per reason for two reasons which **refer directly** to facts on the given profiles

(9)

SEE NEXT PAGE

A	B	C	D
Name	Career chosen	Relevant aspect from profile	How the aspect mentioned in column C will help in this career
Sibongile	IT teacher	(a) IT 75	(a) Does well enough in IT to study to become a teacher
		(b) Team netball OR chairperson of club	(b) Enjoys working with people
		(c) Teaches Bible class	(c) Has some teaching experience; OR this experience will help her to choose if she would like to be teacher
		(d) Chairperson of clubs	(d) Has a talent to organise people, as she will have to do as a teacher
Sibongile	Owner of business which sells PC's	(a) IT 75	(a) Does well enough in IT to learn aspects of IT which she will need in her business
		(b) Accounting and Business Studies	(b) These business oriented subjects will help her in her business
		(c) Team netball OR chairperson of club	(c) Enjoys working with people
		(d) Chairperson of clubs	(d) Has a talent to organise people, as she will have to do in her business
Sibongile	Team programmer	(a) IT 75	(a) IT marks are high enough to study programming for this career
		(b) Team netball OR chairperson of club	(b) Enjoys working with people, so will work well in team
		(c) Chairperson of clubs	(d) Has a talent to organise people, and may have to do so in her programming team
Adam	Individual programmer OR Network technician OR Network administrator	(a) IT 92	(a) High IT marks, can study programming/networking
		(b) Does well in Computer Olympiad	(b) Talent for problem solving – will be good at solving programming/networking problems on his own
		(c) Spends a lot of time on his own	(c) Comfortable with working on his own, as will be required as an individual programmer/network technician/administrator
		(d) Spends time on his own with photography technical work	(d) Self-driven, this will be needed as an individual programmer/network technician/administrator

6.6 Any THREE of the following:

Advantages:

- She can use the Internet to find and study other recipes.
- She can use the Internet to make contact with and swop ideas with other chefs.
- She can use the Internet to sell her recipe books.

Disadvantages:

- There are people who would not buy her books, because they would rather just look up a recipe on the Internet.
- Recipes that she might place on the Internet can be copied and used by others, and she would not receive payment in those cases. (3)

6.7 Any TWO of:

- The progress of the game can be posted on Twitter for everyone to follow.
- It is much simpler than setting up a website to display the results.
- It is quick to update everyone on the game, as a maximum of 140 characters is used with each posting.
- Replacement players can be notified via Twitter (no need for individual SMS's). (2)

6.8

6.8.1 There will be less power use, resulting in less emission of harmful gases. (1)

6.8.2 Any TWO of the following:

- It will not be possible to process statistical data regarding global warming.
- Agencies who work towards the reduction of global warming will not be able to communicate with each other via email.
- It will not be possible educate people regarding global warming via the Internet. (2)

[33]

QUESTION 7

7.1 In the table each person's age is indicated more than once. Each time someone's age is recorded more than once, that is a redundant data element. (2)

7.2 A composite key consists of more than one field. (1)

7.3

7.3.1 The same driver may enter the parking garage more than once, so the driverID will not be a unique field. (1)

7.3.2 driverID, carReg, timestamp (3)
Mark allocation: 1 mark per field

7.4 If 'digitalgal' is deleted from tblUsers, then tblDonations will refer to a record which does not exist. (2)

7.5 count, sum, AS, GROUP BY (each) (8)

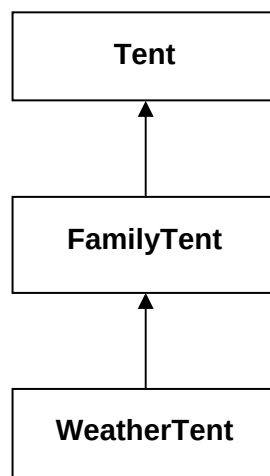
full SQL query:

```
SELECT tblDonations.username, Count(*) AS donations,
      Sum(tblDonations.amount) AS total
FROM tblDonations
GROUP BY tblDonations.username;
```

7.6 Properties/fields of the class: username, status, emailAddress, posts
Methods: incPosts() upgradeStatus()
Subtract one mark per method if method is given a parameter (6)

7.7
7.7.1 Both FamilyTent and WeatherTent have a gazeboArea
WeatherTent could inherit from FamilyTent, so that it is unnecessary to
specify gazeboArea in the class WeatherTent. (3)

7.7.2



(2)

7.8 An access modifier specifies what other classes have access to a field/methods in the class. (2)

7.9 One of:
public, protected, private
(also accept: published, strict private, strict protected) (1)

7.10 When the method needs to return a value. (2)
[33]

180

Total: 180 marks